MA1522 | Midterm Cheet Sheet

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1 Linear Systems

1.1 Introduction to Linear Systems

Linear Equation has Standard Form

$$a_1x_1 + a_2x_2 + ... + a_nx_n = b$$

Where:
- a_1 : Coefficient
- x_1 : Variable
- b : Constant

Sysmtem of Linear Equations

- A Linear system in standard form is written as:

$$\begin{cases} a_{\{11\}}x_1 + a_{\{12\}}x_2 + ... + a_{\{1n\}}x_n = b_1 \\ a_{\{21\}}x_1 + a_{\{22\}}x_2 + ... + a_{\{2n\}}x_n = b_2 \\ ... \\ a_{\{m1\}}x_1 + a_{\{m2\}}x_2 + ... + a_{\{mn\}}x_n = b_m \end{cases}$$

- Homogeneous Linear System $\Leftrightarrow b_1 = b_2 = ... = b_m = 0$
- Solution to Linear System satisfies all equations in the system after substituting the values of variables.
  - General Solution: All possible solutions to the system.
  - Linear System is Consistent $\Leftrightarrow S \notin \emptyset$
  - Linear System is Inconsistent $\Leftrightarrow S \in \emptyset$

1.2 Solving Linear System and Row-Echelon Form

- Augmented Matrix express a linear system in matrix form.

$$\left(\begin{array}{cccc|c} a_{11} & a_{12} & ... & a_1n & b_1 \\ a_{21} & a_{22} & ... & a_2n & b_2 \\ ... & ... & ... & ... & ... \\ a_{\mathrm{m1}} & a_{\mathrm{m2}} & ... & a_{\mathrm{mn}} & b_m \end{array}\right)$$

- Zero Row: A row in which all entries are zero.
- Leading Entry: The first nonzero entry in a nonzero row from left.

Row-Echelon Form (REF)

1. Zero rows are at the bottom of the matrix, if exist.
2. The leading entry of each nonzero row is to the right of the leading entry

Reduced Row-Echelon Form (RREF)

3. Leading entry in each nonzero row is 1.
4. In each pivot column, all other entries, except the leading entry is 0.

- Pivot Column: A column that contains a leading entry.
- Non-Pivot Column: A column that does not contain a leading entry. (May be whole zero column or full of entries)

Number of Solutions from REF

1. No Solution:
   - Leading Entry is in rightmost column.

$$\left(\begin{array}{cc|c} 1 & 0 & 1 \\ 0 & 0 & a \neq 0 \end{array}\right)$$

2. Unique Solution:
   - Every column is a pivot column, except rightmost column.

$$\left(\begin{array}{cc|c} 1 & 0 & b \\ 0 & 1 & b \end{array}\right)$$

3. Infinitely Many Solutions:
   - At least one non-pivot column and no leading entry

$$\left(\begin{array}{cc|c} 1 & 0 & b \\ 0 & 0 & 0 \end{array}\right)$$

1.3 Elementary Row Operations

There are 3 types of ERO:

Operation	Reverse Operation	Condition
$R_i \leftrightarrow R_j$	$R_i \leftrightarrow R_j$	
$R_i = cR_i$	$R_i = \frac{1}{c}R_i$	$c \neq 0$
$R_i = R_i + cR_j$	$R_i = R_i - cR_j$	$i \neq j, c \in \mathbb{R}$

- Row Equivalent Matrices $\Leftrightarrow$ Both linear system have same solution set.

1.4 Gaussian Elimination and Gauss-Jordan Elimination

- Gaussian Elimination(augmented matrix): REF.
  1. Locate the leftmost column that does not consist entirely of zeros.
  2. Interchange the top row with another row, if necessary, to bring a nonzero entry to the top of the column found in Step 1.
  3. For each row below the top row, add a suitable multiple of the top row to it so that the entry below the leading entry of the top row becomes zero.
  4. Now cover the top row in the augmented matrix and begin again with Step 1 applied to the submatrix that remains. Continue this way until the entire matrix is in row-echelon form.
- Gauss-Jordan Elimination(augmented matrix): RREF.
  5. Multiply a suitable constant to each row so that all the leading entries become 1.
  6. Beginning with the last nonzero row and working upward, add suitable multiples of each row to the rows above to introduce zeros above the leading entries.

1.5 More on Linear System

Homogeneous Linear System

- Always consistent
- Always has trivial solution, $x_1 = x_2 = 0$
- Non-trivial solution exists $\Leftrightarrow \geq 1$ non-pivot column in RREF $\Leftrightarrow$ Infinitely many solutions

Linear System with Unknown Variables

- If square matrix, use `det(A)` to quickly check for unique solution (`det(A) eq.not 0`) or edge cases (`det(A) = 0`
- Use ERO to find RREF where:
  - $R_i \leftrightarrow R_j$: Legal
  - $R_i + aR_j$: Legal (Even if $a = 0$)
  - $R_i + \frac{1}{a}R_j$: Illegal if $a = 0$, Check for edge cases
  - $\frac{1}{a}R_i$: Illegal if $a = 0$, Check for edge cases

2 Matrix Algebra

2.1 Definition and Special Types of Matrices

Matrix $\in \mathbb{R}^{m\times n}$, Entries: $a_{ij} \in \mathbb{R}$

$$A = \begin{pmatrix} a_{11} & a_{12} & ... & a_{1n} \\ a_{21} & a_{22} & ... & a_{2n} \\ ... & ... & ... & ... \\ a_{\mathrm{m1}} & a_{\mathrm{m2}} & ... & a_{\mathrm{mn}} \end{pmatrix} = \left(a_{ij}\right)_{m\times n}$$

Special Types of Matrices

- Vector: row vector $\in \mathbb{R}^{1\times n}$; column vector $\in \mathbb{R}^{m\times 1}$
- Zero Matrix $\in 0 \forall a_{ij}$
- Square Matrix $\in m = n$
  - Diagonal Entries: $a_{ii}, i \in \mathbb{N}$
  - Diagonal Matrix: $\in \forall a_{ij} = 0, i \neq j$
  - Scalar Matrix: $\in \forall a_{ij} = \begin{cases} c \text{ if } i=j \\ 0 \text{ if } i\neq j \end{cases}$
  - Identity Matrix: $\in \forall a_{ij} = \begin{cases} 1 \text{ if } i=j \\ 0 \text{ if } i\neq j \end{cases}$

`Identity <: Scalar <: Diagonal <: Square`

Upper Triangle	Strict Upper Triangle	Lower Triangle	Strict Lower Triangle
$\begin{pmatrix} * & * & * \\ 0 & * & * \\ 0 & 0 & * \end{pmatrix}$	$\begin{pmatrix} 0 & * & * \\ 0 & 0 & * \\ 0 & 0 & 0 \end{pmatrix}$	$\begin{pmatrix} * & 0 & 0 \\ * & * & 0 \\ * & * & * \end{pmatrix}$	$\begin{pmatrix} 0 & 0 & 0 \\ * & 0 & 0 \\ * & * & 0 \end{pmatrix}$

2.2 Matrix Algebra

Matrix Addition and Scalar Multiplication

Cummulative	$A + B = B + A$
Associative	$A + (B + C) = (A + B) + C$
Additive Identity	$A + 0 = A$
Additive Inverse	$A + (-A) = 0$
Distributive law	$c(A + B) = cA + cB$
Scalar Addition	$(c + d)A = cA + dA$
Associative	$cdA = c(dA)$
Zero	If $aA = 0_{m\times n}$, $a = 0$ or $A = 0$

Matrix Multiplication

Associative	$A(BC) = (AB)C$
Left Distributive	$A(B + C) = AB + AC$
Right Distributive	$(A + B)C = AC + BC$
Commute with Scalar	$c(AB) = (cA)B = A(cB)$
Multiplicative Identity	$AI = IA = A$
Non-Zero Divisor	$AB = 0_{m\times n}$ does not imply $A = 0$ or $B = 0$
Zero Matrix	$A0 = 0 = 0A$
Power of Matrix	$A^0 = I; A^n = A \times A^{n-1}$

Transpose of Matrix

- $(A^T)^T = A$
- $(A + B)^T = A^T + B^T$
- $(cA)^T = cA^T$
- $(AB)^T = B^TA^T$
- $(ABC)^T = C^TB^TA^T$

To check if a matrix is **symmetric**, check if $A^T = A$

2.3 Linear System and Matrix Equation

Matrix Equation

$$Ax = b = \begin{pmatrix} a_{11} & a_{12} & ... & a_{1n} \\ a_{21} & a_{22} & ... & a_{2n} \\ ... & ... & ... & ... \\ a_{\mathrm{m1}} & a_{\mathrm{m2}} & ... & a_{\mathrm{mn}} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ ... \\ x_n \end{pmatrix} = \begin{pmatrix} b_1 \\ b_2 \\ ... \\ b_m \end{pmatrix}$$

- A : Coefficient Matrix
- x : Variable Vector
- b : Constant Vector

Vector Equation

$$x_1 \begin{pmatrix} a_{11} \\ a_{21} \\ ... \\ a_{\mathrm{m1}} \end{pmatrix} + x_2 \begin{pmatrix} a_{12} \\ a_{22} \\ ... \\ a_{\mathrm{m2}} \end{pmatrix} + ... + x_n \begin{pmatrix} a_{1n} \\ a_{2n} \\ ... \\ a_{\mathrm{mn}} \end{pmatrix} = \begin{pmatrix} b_1 \\ b_2 \\ ... \\ b_m \end{pmatrix}$$

$$= x_1a_1 + x_2a_2 + ... + x_na_n = b$$

- a_i : Column Vector for variable x_i

Block Multiplication

$$\begin{cases} a_{11}x_1 + a_{12}x_2 + ... + a_{1n}x_n = b_1k \\ a_{21}x_1 + a_{22}x_2 + ... + a_{2n}x_n = b_2k \\ ... \\ a_{\mathrm{m1}}x_1 + a_{\mathrm{m2}}x_2 + ... + a_{\mathrm{mn}}x_n = b_{mk} \end{cases}$$

$$= \begin{pmatrix} a_{11} & a_{12} & ... & a_{1n} & b_{11} & b_{12} & ... & b_{1k} \\ a_{21} & a_{22} & ... & a_{2n} & b_{21} & b_{22} & ... & b_{2k} \\ ... & ... & ... & ... & ... & ... & ... & \\ a_{\mathrm{m1}} & a_{\mathrm{m2}} & ... & a_{\mathrm{mn}} & b_{\mathrm{m1}} & b_{\mathrm{m2}} & ... & b_{\mathrm{mk}} \end{pmatrix}$$

When solving block multiplication where coefficient matrix and variable vector has different size, parametrize last few columns as $s_1, s_2 \in \mathbb{R}$

2.4 Inverse of Matrix

- A^{-1} exists $\Leftrightarrow AA^{-1} = I_n = A^{-1}A$
- Non-invertible Matrix: A^{-1} does not exist, Singular Matrix.
- Inverse is unique if exists.

$$A^{-1} = \frac{1}{ad-bc}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix} \text{ for } A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \text{ and } ad - bc \neq 0$$

Cancellation Law for Matrices

- If A^-1 exists:
  - $AX = AY \Rightarrow A^{-1}AX = A^{-1}AY \Rightarrow X = Y$
  - $XA = YA \Rightarrow XAA^{-1} = YAA^{-1} \Rightarrow X = Y$

Finding Inverse of Matrix using ERO

- `rref([A  I]) = [I  A^(-1)]`

Properties of Inverse with Linear System

- If A^{-1} exists
  - $\Leftrightarrow Ax = b$ has unique solution $x = A^{-1}b$
  - $\Leftrightarrow Ax = 0$ has only trivial solution $x = 0$

Properties of Inverse and Equivalence Statement

- The following statements are equivalent:
  1. A^{-1} exists
  2. $(A^{-1})^{-1} = A$
  3. $(A^T)^{-1} = (A^{-1})^T$
  4. $(aA)^{-1} = \frac{1}{a}A^{-1}$
  5. $(AB)^{-1} = B^{-1}A^{-1}$
  6. $A^{-n} = (A^{-1})^n$
  7. `rref([ A I]) = [ I A^(-1)]`
  8. $(A_1A_2A_k)^{-1} = A_k^{-1}A_2^{-1}A_1^{-1}$
  9. $A^{-1} = E_kE_2E_1I$ where E_i are elementary matrices such that $E_kE_2E_1A = I_n$
  10. $Ax = b$ has unique solution for every $b \in \mathbb{R}^n$
  11. $Ax = 0$ has only trivial solution
  12. `det(A) eq.not 0` (if A is square)

Problem Solving Takeaway

- Uses `syms a,b,c,d; AB-BA = 0` to compute general solution for B given A
- Uses Block-Multiplication to find out the right-inverse.

$$(A \;\; I_2) \to (\text{rref}) = \begin{pmatrix} 1 & 0 & 1 & \frac{1}{3} & \frac{2}{3} \\ 0 & 1 & 1 & \frac{1}{3} & -\frac{1}{3} \end{pmatrix}$$

$$B = \begin{pmatrix} \frac{1}{3} - s_1 & \frac{2}{3} - s_2 \\ \frac{1}{3} - s_1 & -\frac{1}{3} - s_2 \\ s_1 & s_2 \end{pmatrix}$$

- For non-square matrix:
  - One-sided inverse (if exists) is not unique as they are parametrized.
  - Have at most one one-sided inverse
  - Left-inverse $\neq$ Right-inverse

- When given two solutions for non-homogenous solution, we may find a solution for homogenous by taking the difference as

$$A(v_1 - v_2) = A(v_1) - A(v_2) = b - b = 0$$
$$\therefore v_1 - v_2 \text{ is solution when b=0}$$

- When given two solutions for homogenous solution, we may find the general solution by (Let A rank 3)

$$S = v_0 + s_1v_1 + s_2v_2 \forall s_1, s_2 \in \mathbb{R}$$
$$A(v_0) = b; A(v_1) = 0; A(v_2) = 0$$

- When given 3 linear system, we can deduce the invertibility of A from B

$$Ax = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}; Ax = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}; Ax = (1; 1; 0)$$

$$Ax = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 0 \end{pmatrix} = B$$

Since B^{-1} exists, A^{-1} exists

2.5 Elementary Matrices

Elementary Matrix: $I \to E$ (by single ERO)

$R_1 + cR_2$	aR_1	$R_1 \leftrightarrow R_2$
$\begin{pmatrix} 1 & 0 \\ c & 1 \end{pmatrix}$	$\begin{pmatrix} a & 0 \\ 0 & 1 \end{pmatrix}$	$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$

$$A \to (r)EA$$
$$A \to (\text{r1}) \to (\text{r2}) ... \to (\text{rk})B$$
$$B = E_k...E_2E_1AA = E_1^{-1}E_2^{-1}...E_k^{-1}B$$

2.6 LU Factorization

- `[L U] = lu(sym(A))`
- $A = LU$
  - L : Lower Triangular Matrix with all diagonal entries = 1
  - U : Upper Triangular Matrix, REF(A)

$$A \to (\text{r1}, \text{r2}, \text{r3}, ..., \text{rk})U$$
$$E_k...E_2E_1A = U$$
$$A = E_1^{-1}E_2^{-1}...E_k^{-1}U$$
$$A = LU$$

- To find L quickly, we may do the following:
  - $E = R_i + cR_j \forall i > j$ (Take top kill bottom)
  - Read off $L = E_1^{-1}E_2^{-1}...E_k^{-1}$
- To solve $Ax = b$ using LU Factorization:

$$Ax = LUx = b$$
$$\text{Let } y = Ux$$
$$Ly = b$$
$$Ux = y$$

Problem Solving Takeaway

- Once L is computed from the Coefficient Matrix, we can use that to multiply with B and solves the LS easily. $Ax = b; LUx = Lb; Ux = b$ (PS4, 2b)
- If A is invertible, L and U is invertible too. (PS4, 1a)
- If A is LU-factorable, it doesnt say anything about invertibility of A (T3, E1a)
- If A is not LU-factorable, there exists a permutation matrix, $P_{R_i \leftrightarrow R_j}$ such that PA is LU-factorable. (T3, E1b)

2.7 Determinant

By Cofactor Expansion

- **Only defined for square matrix**.

$$\det(A) = a_{11}A_{il} + a_{12}A_{12} + ... + a_{1n}A_{1n} = \sum_{k=1}^{n} a_{ik}A_{ik}$$
$$\text{where } A_{ik} = (-1)^{i+k}\det(M_{ik})$$

- A_{ik} : Cofactor of entry a_{ik}
- M_{ik} : Minor Matrix, delete i-th row and k-th column of A
- Determinant of transpose: $\det(A^T) = \det(A)$
- Determinant of Triangular Matrix: Product of diagonal entries.

By Reduction

- Every Elementary Matrix has known determinant:

$$\det\left(E_{R_i + cR_j}\right) = 1$$
$$\det\left(E_{aR_i}\right) = a$$
$$\det\left(E_{R_i \leftrightarrow R_j}\right) = -1$$

- Let R be RREF of A after k EROs, then

$$\det(R) = d_1d_2...d_k \text{ since R is upper triangular}$$
$$= \det(E_k)...\det(E_2)\det(E_1)\det(A)$$
$$\therefore \det(A) = d_1d_2...\frac{d_k}{\det(E_1)\det(E_2)...\det(E_k)}$$

Properties of Determinant

$$\det(AB) = \det(A)\det(B)$$
$$\det(A^{-1}) = \frac{1}{\det(A)}$$
$$\det(A^T) = \det(A)$$
$$\det(cA) = c^n\det(A) \text{ for } A \in \mathbb{R}^{n \times n}$$

Deeper Understanding of Determinant

- If two rows / columns are the same, the matrix is not invertible

$$\text{Let } A = \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix}$$

Performing ERO on A, det(A) changes by (1),(-1),(a)

IF a=d, e=b, c=f:

$$A = \begin{pmatrix} a & b & c \\ a & b & c \\ g & h & i \end{pmatrix} \to \begin{pmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}\begin{pmatrix} a & b & c \\ 0 & 0 & 0 \\ g & h & i \end{pmatrix} = A'$$
$$\det(A) = \det\left(E_{R_2 - R_1}\right) * \det(A') = 1 * 0 = 0$$
$$\therefore \det(A) \text{ can be derived through ERO}$$

2.8 Adjoint

- Adjoint of Matrix A is $\text{adj}(A) = (A_{ji})^T_{n \times n}$

$$\text{adj}(A) = \begin{pmatrix} A_{11} & A_{12} & ... & A_{1n} \\ A_{21} & A_{22} & ... & A_{2n} \\ ... & ... & ... & ... \\ A_{n1} & A_{n2} & ... & A_{nn} \end{pmatrix}^T$$
$$A_{ij} : \text{Cofactor of entry } a(ij)$$
$$A \times \text{adj}(A) = \det(A)I$$
$$A^{-1} = \frac{1}{\det(A)}\text{adj}(A) \text{ if} \det(A) \neq 0$$

Cramer's Rule

- If A^{-1} exists, the unique solution to $Ax = b$ is given by:

$$x = A^{-1}b$$
$$= \frac{1}{\det(A)}\text{adj}(A)b$$
$$= \frac{1}{\det(A)}\begin{pmatrix} \det(A_1(b)) \\ \det(A_2(b)) \\ ... \\ \det\left(A_{n(b)}\right) \end{pmatrix}$$

(T3, 7)

Problem Solving Takeaway

- If two rows/columns are the same, the adjoint is not invertible. Prove with matlab (PS3, 3b).
- If A is not invertible, $\text{adj}(A)$ is the solution to homogeneous solution $Ax = 0$ as $A \times \text{adj}(A) = \det(A)I = 0$ (PS3, 3c)

3 Euclidean Vector Space

3.1 Euclidean Vector Spaces

$$\mathbb{R}^n = \left\{ v = \begin{pmatrix} x_1 \\ x_2 \\ ... \\ x_n \end{pmatrix} \mid x_i \in \mathbb{R} \right\}$$

- $\mathbb{R}^n$ Euclidean N-space
- $v = (x_1, x_2, ..., x_n)$: Vector in $\mathbb{R}^n$
- All matrix algebraic operation is valid in vector

3.2 Dot Product, Norm, Distance

- **Dot-Product:**

$$u.v = u_1v_1 + u_2v_2 + ... + u_nv_n = \sum_{i=1}^{n} u_iv_i = u^Tv$$

- **Norm / Length / Magnitude:**

$$\|u\| = \sqrt{u.u} = \sqrt{u_1^2 + u_2^2 + ... + u_n^2}$$

- Properties of **Inner Product and Norm**:

$$u.v = v.u$$
$$cu.v = (cu).v = u.(cv)$$
$$u.(av + bw) = a(u.v) + b(u.w)$$
$$u.u \geq 0 \text{ and } u.u = 0 \Leftrightarrow u = 0$$
$$\|cu\| = |c|\ \|u\|$$

- **Unit Vector**

$$\|u\| = 1$$
$$\hat{u} = \frac{u}{\|u\|}$$

- Distance between two vectors

$$d(u,v) = \|u - v\| = \sqrt{(u_1 - v_1)^2 + (u_2 - v_2)^2 + ... + (u_n - v_n)^2}$$
$$= \sqrt{(u - v).(u - v)}$$

- Angle between two vectors

$$\cos(\theta) = \frac{u.v}{\|u\|\ \|v\|}$$
$$\theta = \arccos\left(\frac{u.v}{\|u\|\ \|v\|}\right)$$
$$0 \leq \theta \leq \pi$$

3.3 Linear Combination and Linear Spans

- Geometrically, Span $\text{span}\{u_1, u_2, ...\}$ is all the position that one can reach by taking a linear combination of u_1, u_2 where $\{c_1u_1 + c_2u_2 \mid c_1, c_2 \in \mathbb{R}\}$

$$\text{span}\{u_1, u_2, ..., u_k\} = \{c_1u_1 + c_2u_2 + ... + c_ku_k \mid c_1, c_2, ..., c_k \in \mathbb{R}\}$$

- Check if Vector $v \in \text{span}(S)$

$$\text{span}\{u_1, u_2, ...\} \Leftrightarrow v = c_1u_1 + c_2u_2 + ... + c_ku_k$$
$$\Leftrightarrow Ax = v \text{ is consistent}$$

- To check if span(S) is the whole euclidean n-space:

$$\text{span}(S) = R^n$$
$$\Leftrightarrow \text{rref(S) has non-zero rows}$$
$$\Leftrightarrow Ax = v \text{ is consistent}$$

(PS5, 2)

Properties of Linear Span

span(S) has following properties:
1. $0 \in \text{span}(S)$
2. $u + v \in \text{span}(S)$
3. $\alpha u \in \text{span}(S)$

Equivalence of Span

- To check if linear spans are equal by:

$$\text{Let } S = \{v_1, v_2\}, T = \{u_1, u_2\}$$
$$\text{span}(S) \subseteq \text{span}(T) \Leftrightarrow (u_1u_2 \mid v_1 \mid v_2) \text{ is consistent}$$
$$\text{span}(S) = \text{span}(T) \Leftrightarrow \text{span}(S) \subseteq \text{span}(T) \wedge \text{span}(T) \subseteq \text{span}(S)$$

3.4 Subspaces

- Subspace is a subset of a euclidean-vector space that is itself a vector space under the same operations of addition and scalar multiplication as the larger space. (Similar idea with linear span)

Let V be Solution Set to linear system which can be express:
- Implicitly (use to test if set/span in V)

$$V = \{u \in \mathbb{R}^n \mid Au = b\}$$

- Explicitly (use to test if V in span)

$$V = \{u + s_1v_1 + s_2v_2 \mid s_1, s_2 \in \mathbb{R}\}$$

Subspace

$V \subseteq \mathbb{R}^n$ is subspace if:
1. $0 \in V$
2. For any $v \in V$, $\alpha v \in V$
3. For any $u, v \in V$, $u + v \in V$

OR (Either one)
- Find linear span where $V = \text{span}\{v_1, v_2\}$
- Solution set, S for LS $Au = 0$ (Homogenous Linear System)

Subspace of R^2

1. Zero space
2. Lines
3. Whole $\mathbb{R}^2$

Subspace of R^3

1. Zero space
2. Lines
3. Planes
4. Whole $\mathbb{R}^3$

Affine Space

$$W = \{u + v \mid A(u + v) = b\} = u + v := \{u + v \mid Av = 0\}$$

Problem Solving Takeaway

- When seeing a Solution Set with unparametrized vector, we cannot determine if V is not a subspace until we are sure that the unparametrized vector is not linearly dependent on the other vectors.

$$V = \left\{ \begin{pmatrix} 4 \\ -3 \\ 9 \\ 1 \end{pmatrix} + s_1 \begin{pmatrix} 6 \\ 6 \\ 8 \\ 2 \end{pmatrix} + s_2 \begin{pmatrix} 6 \\ 3 \\ 7 \\ 1 \end{pmatrix} + s_3 \begin{pmatrix} -2 \\ 3 \\ -4 \\ 0 \end{pmatrix} \mid s_1, s_2, s_3 \in \mathbb{R} \right\}$$

$$\text{BUT rref}(u_1, u_2, u_3, v) = \begin{pmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & -2 \end{pmatrix}$$

$$\therefore v = u_1 - u_2 - 2u_3 \text{ and } V \text{ is subspace}$$

- If an implicit solution set cannot be reduced to linear system, it is not subspace. (T4,6f)

$$S = \left\{ \begin{pmatrix} a \\ b \\ c \\ d \end{pmatrix} \mid ab = cd \right\}$$

$$\begin{pmatrix} 1 \\ 0 \\ 1 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 1 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ 2 \\ 0 \end{pmatrix} \notin S$$

4 Equivalent Statements

4.1 Invertibility

- A^{-1} exists
- A^T is invertible
- (left inverse), There exists B such that $BA = I$
- (right inverse), here exists B such that $AB = I$
- RREF of A is I
- A is expressed as product of elementary matrices, $A = E_1^{-1}E_2^{-1}E_k^{-1}I$
- $Ax = 0$ has only trivial solution
- $Ax = b$ has unique solution
- $\det(A) \neq 0$